

Property Flags aur Descriptors

Object ki property sirf key-value nahi hoti. Value ke saath 3 **flags** bhi hote hain:

Flag	<code>true</code> ho to	<code>false</code> ho to
<code>writable</code>	value badal sakte ho	read-only
<code>enumerable</code>	loops me dikhti hai	loops me nahi dikhti
<code>configurable</code>	delete kar sakte ho aur flags badal sakte ho	delete/flags change nahi

Normal tareeke se banayi property me teeno `true` hote hain.

Flags dekhna

```
javascript
let user = { name: "John" };
Object.getOwnPropertyDescriptor(user, "name");
// { value: "John", writable: true, enumerable: true, configurable: true }
```

Flags badalna / property banana

```
javascript
Object.defineProperty(obj, "name", descriptor);
```

- Property pehle se hai to flags update hote hain.
- Nayi property banti hai to jo flag nahi diya wo `false` maana jaata hai.

Non-writable (read-only)

```
javascript
Object.defineProperty(user, "name", { writable: false });
user.name = "Pete"; // strict mode me error, warna chupchaap ignore
```

Non-enumerable

`for...in` aur `Object.keys` me nahi dikhti. Built-in `toString` isi wajah se loop me nahi aata.

```
javascript
```

```
Object.defineProperty(user, "toString", { enumerable: false });
```

Non-configurable

- Property delete nahi ho sakti aur flags nahi badal sakte.
- Ye **ek tarfa raasta** hai. Wapas `true` nahi kar sakte.
- Value phir bhi badal sakti hai agar `writable: true` ho.
- Ek hi exception: `writable` ko `true` se `false` kar sakte ho.

Example: `Math.PI` teeno flags `false` ke saath hai, isliye use koi badal nahi sakta.

Poori tarah "constant" banane ke liye:

javascript

```
Object.defineProperty(user, "name", { writable: false, configurable: false });
```

Ek saath kai properties

javascript

```
Object.defineProperties(user, {  
  name: { value: "John", writable: false },  
  surname: { value: "Smith", writable: false }  
});
```

Saare descriptors ek saath / flags-wali cloning

javascript

```
Object.getOwnPropertyDescriptors(obj);  
  
let clone = Object.defineProperties({}, Object.getOwnPropertyDescriptors(obj));
```

Normal copy (`clone[key] = obj[key]`) flags aur symbol/non-enumerable properties copy nahi karti. Ye tareeka karta hai.

Poore object ko lock karna

Method	Kya rokta hai
<code>Object.preventExtensions(obj)</code>	nayi property jodna
<code>Object.seal(obj)</code>	jodna aur hatana (<code>configurable: false</code>)
<code>Object.freeze(obj)</code>	jodna, hatana aur badalna (<code>configurable: false</code> , <code>writable: false</code>)

Check karne ke liye: `Object.isExtensible`, `Object.isSealed`, `Object.isFrozen`. Ye practice me kam use hote hain.

Yaad rakho

- Property = value + 3 flags.
- `defineProperty` se flags badalte hain (nayi property me default `false`).
- `configurable: false` wapas nahi hota.
- `freeze` sabse strict hai.